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TERM END EXAMINATIONS (TEE) – October 2026

Programme	: **B.Tech**		**Semester**	:	**Fall Semester 2026-27**
Course Name	: **Computer Architecture and Organization**		**Course Code**	:	**CSE2003**
Date/Session	: **14.10.2026 S-2**		**Slot**	:	**C14+E11+E12**
Time	: **3 Hrs.**		**Max. Marks**	:	**100**

Answer ALL the Questions

Q. No.		Question Description	Marks
		PART A – (60 Marks)	
1	(a)	Considering a basic functional blocks in a computer, which architecture is best suited for real world applications? Discuss the pros and cons of each architecture. A computer has a byte addressable memory unit containing 128K words of 32 bits each. The computer has an instruction format with 4 fields: i.e. an op-code field, a mode field to specify 1 of 7 addressing modes, a register address field to specify 1 of 60 registers and a memory address field. If instruction size is 32 bits long, then determine the number of possible operations that can be represented with above instruction format. Show instruction format for the given case.	12
		OR	
	(b)	Briefly discuss the addressing modes for the following given syntax: (i).R1← Operand (ii).R2← M[Address] (iii).R3←M[M[Address]] (iv).R4← R5 (v).R6←M[R7] (vi). M[Address]← R8,M[Address]	12
2	(a)	Perform Booth's algorithm for the binary multiplication of the given numbers (-15) and (6). Illustrate step wise calculation showing the register contents clearly for each iteration.	12
		OR	
	(b)	Consider the following 14-bit floating-point number representation format. The first bit is the sign of the number. The next five bits represent exponent values, which has an implied base of 2. The last 6 bits represent the fractional part of the mantissa. Perform Subtraction and Multiplication operation for the given operands. Specify all the steps clearly while performing the operations. A= B=	12

0	10001	011011
1	01111	101010

3	(a)	Apply Direct mapping, Associative mapping and (4-way)Set associative mapping for cache of size 16KB with block size of 256 bytes. Determine how the main memory addresses are mapped to the cache lines assuming the size of the main memory is 128KB. Show the mapping process clearly with suitable diagrams and expressions.	12
		OR	
	(b)	Consider a machine with a byte addressable main memory of 64KB and block size of 8 bytes. Consider a cache consisting of 32 lines is used with this machine. **i).** Assume a direct mapped cache, show the address format and the division of main memory address into tag, line number, and offset. **ii).** Assume a four-way set-associative cache, show the address format and determine the number of blocks in main memory, number of sets in cache memory and size of tag. **iii).** For the main memory addresses (in hexadecimal) of F001 and CABB, give the corresponding tag, line number and offset values for a direct mapped cache **iv).** For the main memory addresses (in hexadecimal) of 012C and ABBE, give the corresponding tag and offset values for a fully-associative cache.	12
4	(a)	Given a sequence of instructions in a pipelined processor, identify and illustrate the different types of data hazards with suitable examples. Analyse a disk access operation and investigate the different components contributing to the total disk access time.	12
		OR	
	(b)	Explicate the concept of address translation mechanism used in virtual memory. A DMA controller transfers 16 bit words to memory using cycle stealing. The word are assembled from a device that transmit character at the rate of 2400 character/ sec, the CPU is fetching instruction at an average rate of 1 million instructions /sec, by how much will the CPU be slow down because of DMA transfer?	12
5	(a)	Illustrate the operational sequence of Programmed I/O and Interrupt-driven I/O for transferring data between an I/O device and the processor, using a suitable flowchart or sequence diagram. For a system requiring high-speed data transfer between an I/O device and main memory, which mode will be preferred? Justify with suitable reasons.	12
		OR	
	(b)	Discuss different types of Interconnection architectures employed in multi-processor systems Consider an instruction pipeline having 5 stages FI (Fetch Instruction), DI (Decode Instruction), FO (Fetch Operand), EI (Execute Instruction) and WO (Write Operand) with duration 60, 50, 70, 85 and 80 ns. Given latch delay is 15 ns. Then compute speedup ratio and Pipeline time for 1000 tasks.	12
		PART B – (40 Marks)	
6		Categorize different types of registers based on their function in a computer system. Demonstrate the role of each of the following registers with a suitable example.	8
7		Illustrate the step-by-step process to obtain the floating point representation of the number 43.225 using single precision format. Also show IEEE 32-bit and 64-bit floating point representation clearly.	8
8		Consider a 4-way set associative mapping with 16 cache blocks. The memory block requests are in the order- 0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155,48,32,8,92 If LRU replacement policy is used, which even numbered cache blocks are not available in the cache? Also, calculate the hit ratio and miss ratio.	8

9		Given a data-transfer scenario between a peripheral device and main memory, apply the concept of Direct Memory Access (DMA) to demonstrate the data-transfer sequence with a suitable sketch, and illustrate the operation of different DMA transfer methods.	8
10		Analyse Flynn's taxonomy to classify computer architectures based on the multiplicity of instruction and data streams. Demonstrate each of the following categories with a suitable example and diagram:	8
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